

# The Kasami Permutation Criterion — a refactored Lean 4 library

Module map, dependency DAG, and Dobbertin (1999) correspondence,  
organised by mathematical role rather than by paper numbering

Formalisation library `KasamiPermutations.*`

## Abstract

This document describes the *refactored* Lean 4 library `KasamiPermutations.*`, which formalises — from the `Mathlib` foundations, end to end and `sorry`-free — the opening of H. Dobbertin’s *Theorem 1 (Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets*, 1999, pp. 133–138): equation (1), the permutation criterion for the generalized Kasami map  $q_\alpha$ , and the first substantive step  $(1) \Rightarrow (2)$ . The refactoring reorganises the modules *by mathematical role* (a Frobenius theory, a trace theory, a Mersenne number-theory fact, the Kasami-map definitions, and two permutation criteria) rather than by their numbering in the paper, following the “small, coherent, reusable module” principle. We give the module dependency DAG, the old  $\rightarrow$  new rename map, and a declaration-level Dobbertin  $\leftrightarrow$  Lean correspondence, and we record the simplifications made. Every headline result rests only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

## Contents

|          |                                                              |          |
|----------|--------------------------------------------------------------|----------|
| <b>1</b> | <b>The refactoring at a glance</b>                           | <b>2</b> |
| <b>2</b> | <b>Old <math>\rightarrow</math> new rename map</b>           | <b>2</b> |
| <b>3</b> | <b>Module dependency DAG</b>                                 | <b>3</b> |
| <b>4</b> | <b>The foundation layer: finite-field and number theory</b>  | <b>3</b> |
| 4.1      | FiniteField/Frobenius — the Frobenius endomorphism . . . . . | 3        |
| 4.2      | FiniteField/Trace — the additive (absolute) trace . . . . .  | 4        |
| 4.3      | NumberTheory/MersenneNumbers . . . . .                       | 4        |
| <b>5</b> | <b>The Kasami map and its special values</b>                 | <b>4</b> |
| 5.1      | KasamiMap — the definitions . . . . .                        | 4        |
| 5.2      | SpecialValues — the “only if” direction . . . . .            | 5        |
| <b>6</b> | <b>The two permutation criteria</b>                          | <b>5</b> |
| 6.1      | TraceFreeCriterion (paper: Theorem 5) . . . . .              | 5        |
| 6.2      | TraceVersionCriterion (paper: Theorem 8) . . . . .           | 5        |
| <b>7</b> | <b>The headline: PermutationCriterion</b>                    | <b>6</b> |
| 7.1      | The permutation criterion for $q_\alpha$ . . . . .           | 6        |
| 7.2      | The step $(1) \Rightarrow (2)$ . . . . .                     | 6        |
| 7.3      | The root count (two cases) . . . . .                         | 6        |
| <b>8</b> | <b>What the whole development reduces to</b>                 | <b>6</b> |

## 1 The refactoring at a glance

The library proves, for  $L = \mathbb{F}_{2^n}$  with  $\gcd(k, n) = 1$ ,  $k < n$  and  $kk' \equiv 1 \pmod{n}$ , that the generalized Kasami map

$$q_\alpha(z) = \left( \sum_{i=1}^{k'} z^{2^{ik}} + \alpha \operatorname{Tr}(z) \right) \cdot z^{(2^n-1)-(2^k+1)} \quad (\alpha \in \{0, 1\})$$

is a permutation of  $L$  iff  $k' + \alpha n \equiv 1 \pmod{2}$ , and that equation (1),

$$cx^{2^k+1} = \sum_{i=1}^{k'} x^{2^{ik}} + \alpha \operatorname{Tr}(x), \quad (1)$$

implies the linearized equation (2)  $\ell(x) = c^{2^k} x^{2^{2k}} + x^{2^k} + cx + 1 = 0$ .

**Organizing principle.** The previous layout named files and namespaces after the paper’s numbering (**Theorem5**, **Theorem8Trace**, **Theorem8C1**, **Q1General**, **Equation1**, **Defs**, **Setup**, **FiniteFieldPrereqs**). The refactored layout instead groups declarations by *what they are about*, so that each module is a small, self-describing, independently reusable theory:

- a **Frobenius** theory and a **trace** theory of a finite field of characteristic 2 (upstreamable, Kasami-agnostic);
- a **Mersenne number-theory** fact (upstreamable, field-agnostic);
- the **Kasami-map definitions** and their **special values**;
- the two **permutation criteria** — one for the constant-bit map  $q^{(\varepsilon)}$ , one for the trace map  $g = q_1$  — and the **headline** assembly with the  $(1) \Rightarrow (2)$  step.

## 2 Old $\rightarrow$ new rename map

| Old file / namespace                                                                                      | New file / namespace                                                   | Mathematical role                                                       |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Defs.lean</b>                                                                                          | <b>KasamiMap.lean</b>                                                  | definitions of $\operatorname{Tr}$ , $q_\alpha$ , eqn (1), $\ell$ , $Q$ |
| <b>Dobbertin1999.Paper</b>                                                                                | <b>Kasami</b>                                                          |                                                                         |
| <b>Setup.lean</b> (values)                                                                                | <b>SpecialValues.lean</b><br><b>Kasami</b>                             | $q_\alpha(0)$ , $q_\alpha(1)$ ; the “only if” direction                 |
| <b>Setup.lean</b> (arithmetic)                                                                            | <b>NumberTheory/MersenneNumbers.lean</b><br><b>Kasami.NumberTheory</b> | image of $z \mapsto z^{2^k+1}$ and modular inverse                      |
| <b>FiniteFieldPrereqs.lean</b> (Frobenius)                                                                | <b>FiniteField/Frobenius.lean</b><br><b>Kasami.FiniteField</b>         | Frobenius $x \mapsto x^{p^r}$ facts                                     |
| <b>FiniteFieldPrereqs.lean</b> (trace)<br>+ <b>Theorem8Trace.lean</b><br>+ <b>Theorem8C1.lean</b> (trace) | <b>FiniteField/Trace.lean</b><br><b>Kasami.FiniteField</b>             | additive/absolute trace and its identities                              |
| <b>Theorem5.lean</b><br><b>Dobbertin.Thm5</b>                                                             | <b>TraceFreeCriterion.lean</b><br><b>Kasami.TraceFreeCriterion</b>     | criterion for the constant-bit map $q^{(\varepsilon)}$                  |

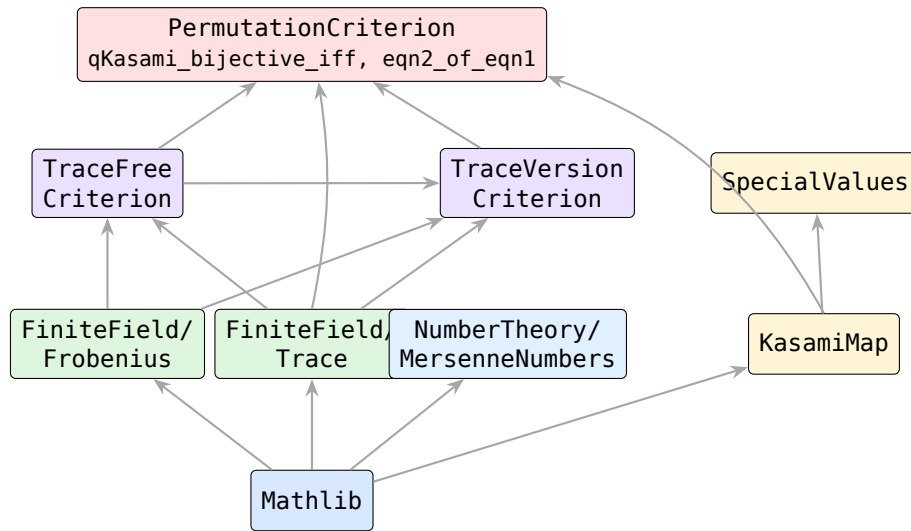
| Old file / namespace                                                 | New file / namespace                                       | Mathematical role                                            |
|----------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| Q1General.lean<br>Dobbertin.Thm8C1Gen<br>+ Theorem8C1.lean<br>(gmap) | TraceVersionCriterion.lean<br>Kasami.TraceVersionCriterion | criterion for the trace map $q_1$                            |
| Equation1/Equation1.lean<br>Dobbertin1999.Paper                      | PermutationCriterion.lean<br>Kasami                        | headline criterion for $q_\alpha$ ;<br>(1) $\Rightarrow$ (2) |

The headline declarations were also renamed to describe the pattern they prove:

| Old name           | New name                 | Statement                                                    |
|--------------------|--------------------------|--------------------------------------------------------------|
| theorem_5          | qeps_bijective_iff       | $q^{(\varepsilon)}$ bijective $\iff \varepsilon \equiv k'+1$ |
| theorem_1          | qKasami_bijective_iff    | $q_\alpha$ bijective $\iff k'+\alpha n$ odd                  |
| theorem_1_case1    | linearized_root_unique   | $\ell(x) = 0$ has a unique root (case 1)                     |
| theorem_1_case2    | eqn1_nonzero_root_unique | eqn (1) has a unique nonzero root (case 2)                   |
| gmap_bijective_iff | gmap_bijective_iff       | $g = q_1$ bijective $\iff k'+n$ odd                          |
| eqn2_of_eqn1       | eqn2_of_eqn1             | (1) $\Rightarrow$ (2)                                        |

### 3 Module dependency DAG

Every import points “upward” in the DAG below; reading the modules in topological order is the same as reading the mathematics from the foundations up. The number-theory fact and the special values form the small “opening/only-if” branch that is independent of the main root-count argument.



Colour key: Mathlib finite-field theory number theory Kasami definitions criteria headline.

## 4 The foundation layer: finite-field and number theory

### 4.1 FiniteField/Frobenius — the Frobenius endomorphism

A Kasami-agnostic theory of the map  $x \mapsto x^{p^r}$  on a finite field.

```

namespace Kasami.FiniteField
lemma frob_cycle (x : F) : x ^ Fintype.card F = x
lemma frob_mod (hn : Fintype.card F = p ^ n) (x : F) (r : ℕ) :
  x ^ (p ^ r) = x ^ (p ^ (r % n))
lemma frob_bijective (r : ℕ) : Function.Bijective (fun x : F => x ^ (p ^ r))
lemma pow_field_bijective (ha : Nat.Coprime (Fintype.card F - 1) a) (ha_pos : 0 < a)
  :
  Function.Bijective (fun x : F => x ^ a)

```

These package Fermat’s little theorem (`FiniteField.pow_card`), Frobenius injectivity (`iterateFrobenius`) and the pigeonhole “injective  $\Rightarrow$  surjective” device (`Finite.injective_iff_surjective`).

## 4.2 FiniteField/Trace — the additive (absolute) trace

**Dobbertin (1999) — p. 135, trace properties**

$\text{Tr}(x) = \text{Tr}(x^2)$  and  $\text{Tr}(x) \in \{0, 1\}$ ; the absolute trace is  $\mathbb{F}_2$ -linear.

```

namespace Kasami.FiniteField
def truncTrace (m : ℕ) (x : F) : F := ∑ i ∈ Finset.range m, x ^ (2 ^ i)
lemma truncTrace_add : truncTrace m (x + y) = truncTrace m x + truncTrace m y
lemma truncTrace_sq_add_self : truncTrace m x ^ 2 + truncTrace m x = x ^ (2 ^ m) + x
theorem trace_frob_shift (hn) (k) (x) : truncTrace n (x ^ (2 ^ k)) = truncTrace n x
theorem trace_artin_schreier_zero (hn) (k) (t) : truncTrace n (t ^ (2 ^ k) + t) = 0
theorem trace_sq (hn) (x) : (truncTrace n x) ^ 2 = truncTrace n x
theorem trace_bit (hn) (x) : truncTrace n x = 0 ∨ truncTrace n x = 1

```

**Remark.** This single module now collects trace facts that were previously scattered across three files (`FiniteFieldPrereqs`, `Theorem8Trace`, `Theorem8C1`). `truncTrace_add` is the “freshman’s dream” (`add_pow_char_pow`); `truncTrace_sq_add_self` is the telescoping identity; `trace_frob_shift`/`trace_artin_schreier_zero` are the paper’s Frobenius invariance and Artin–Schreier vanishing; and `trace_sq`/`trace_bit` express  $\text{Tr}(x) \in \{0, 1\}$ .

## 4.3 NumberTheory/MersenneNumbers

**Dobbertin (1999) — p. 135, opening of the proof**

First note that  $(2^k - 1)^{-1} \pmod{2^n - 1}$  exists, since  $\text{gcd}(k, n) = 1$ .

```

namespace Kasami.NumberTheory
theorem mersenne_coprime (h : Nat.Coprime k n) :
  Nat.Coprime (2 ^ k - 1) (2 ^ n - 1)
theorem inv_mod_exists (h : Nat.Coprime k n) (hn : 1 < 2 ^ n - 1) :
  ∃ b, (2 ^ k - 1) * b % (2 ^ n - 1) = 1

```

The key Mathlib input is  $\text{gcd}(2^k - 1, 2^n - 1) = 2^{\text{gcd}(k, n)} - 1$  (`Nat.pow_sub_one_gcd_pow_sub_one`).

# 5 The Kasami map and its special values

## 5.1 KasamiMap — the definitions

```

namespace Kasami
def Tr (n : ℕ) (x : L) : L := ∑ i ∈ Finset.range n, x ^ (2 ^ i)
def qKasami (n k k' α : ℕ) (z : L) : L :=
  ∑ (( i ∈ Finset.Icc 1 k', z ^ (2 ^ (i * k))) + α( : L) * Tr n z)
  * z ^ (2 ^ n - 1 - (2 ^ k + 1))

```

```

def eqn1 (n k k' α : ℕ) (c x : L) : Prop :=
  c * x ^ (2 ^ k + 1) = ∑( i ∈ Finset.Icc 1 k', x ^ (2 ^ (i * k))) + α( : L) * Tr n
  x
def ell (k : ℕ) (c x : L) : L := c ^ (2 ^ k) * x ^ (2 ^ (2 * k)) + x ^ (2 ^ k) + c
  * x + 1
def Qmap (k : ℕ) (c γ x : L) : L := c * x ^ (2 ^ k) + γ ^ 2 * x + γ

```

## 5.2 SpecialValues — the “only if” direction

**Dobbertin (1999) — p. 135**

*We have  $q_\alpha(0) = 0$ ; and  $k' + \alpha n \equiv 0 \pmod{2}$  is equivalent to  $q_\alpha(1) = k' \cdot 1 + \alpha \text{Tr}(1) = 0$ .*

```

namespace Kasami
theorem qKasami_zero α() : qKasami n k k' α 0 = 0
theorem qKasami_one α() : qKasami n k k' α 1 = ((k' + α * n : ℕ) : L)
theorem qKasami_one_eq_zero_iff α() :
  qKasami n k k' α 1 = 0 ↔ (k' + α * n) % 2 = 0

```

## 6 The two permutation criteria

### 6.1 TraceFreeCriterion (paper: Theorem 5)

**Dobbertin (1999) — Theorem 5, p. 137**

*$q^{(\varepsilon)}$  is a permutation polynomial on  $L$  if and only if  $\varepsilon \equiv k' + 1 \pmod{2}$ .*

```

namespace Kasami.TraceFreeCriterion
def qeps (n k k' : ℕ) ε( x : F) : F := (sTrace k k' x + ε) * x ^ (2 ^ n - 1 - (2 ^ k
  + 1))
theorem qeps_bijective_iff (hn) (hk0) (hkn) (hcop) (hkk') (hexp) (εh : ε = 0 ∨ ε =
  1) :
  Function.Bijective (qeps n k k' ε) ↔ ε( = 1 ↔ Even k')

```

The proof reduces  $q^{(\varepsilon)}(x) = c$  (for a fixed value  $c$ ) to the linearized equation  $\ell(x) = 0$  by adding to it its  $2^k$ -th power (`ell_of_eq`), and then counts roots via a case split (`root_count`, `Q_factor`, `ell0_root_imp_image`).

### 6.2 TraceVersionCriterion (paper: Theorem 8)

Here the constant bit  $\varepsilon$  is replaced by the actual trace  $\text{Tr}(x)$ , giving the map  $g(x) = q^{(\text{Tr} x)}(x)$ , i.e. the  $\alpha = 1$  object  $q_1$ .

```

namespace Kasami.TraceVersionCriterion
noncomputable def gmap (n k kk : ℕ) (x : F) : F := qeps n k kk (truncTrace n x) x
theorem gmap_bijective_iff (hn) (hk0) (hkn) (hcop) (hkk') (hexp) :
  Function.Bijective (gmap n k kk) ↔ Odd (kk + n)

```

The three admissible parities of  $(k', n)$  (they are never both even, since  $\text{gcd}(k', n) = 1$ ) are handled uniformly through the single criterion  $\text{Odd}(k' + n)$ ; the proof mirrors the trace-free root count with the trace now carried along by `trace_artin_schreier_zero` and `trace_bit`.

## 7 The headline: **PermutationCriterion**

### 7.1 The permutation criterion for $q_\alpha$

**Dobbertin (1999) — Theorem 1 (first part), p. 135**

$q_\alpha$  is a permutation of  $L = \mathbb{F}_{2^n}$  if and only if  $k' + \alpha n \equiv 1 \pmod{2}$ .

```
namespace Kasami
theorem qKasami_bijective_iff (hn) (hk) (hcop) (hk') (hk0) (hexp)
  α() (αh : α = 0 ∨ α = 1) :
  Function.Bijective (qKasami n k k' α) ↔ (k' + α * n) % 2 = 1
```

The two bridge lemmas `qKasami_zero_eq_qeps` and `qKasami_one_eq_gmap` identify the paper's  $q_\alpha$  with the library objects `qeps` (for  $\alpha = 0$ ) and `gmap` (for  $\alpha = 1$ ), so that `qKasami_bijective_iff` is assembled directly from `qeps_bijective_iff` and `gmap_bijective_iff`:

$$q^{(\varepsilon)} \text{ criterion} \xrightarrow{\alpha=0} q_\alpha \text{ criterion} \xleftarrow{\alpha=1} g = q_1 \text{ criterion}$$

(In Lean: `qeps_bijective_iff` and `gmap_bijective_iff` assemble `qKasami_bijective_iff`.)

### 7.2 The step (1) $\Rightarrow$ (2)

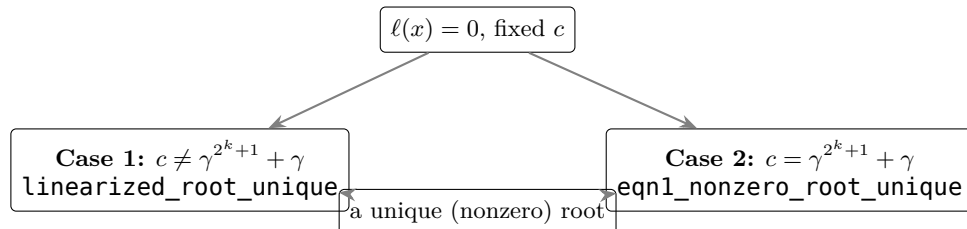
**Dobbertin (1999) — p. 135–136**

Adding to (1) its  $2^k$ -th power yields  $\ell(x) = c^{2^k} x^{2^{2k}} + x^{2^k} + cx + 1 = 0$ . (2)

```
namespace Kasami
theorem eqn2_of_eqn1 (hn) (hkk1 : k * k' % n = 1) α() (c x : L)
  (hx : x ≠ 0) (h : eqn1 n k k' α c x) : ell k c x = 0
```

**Correction 1.** The naive statement is *false* at  $x = 0$ : there the cleared equation (1) holds vacuously ( $0 = 0$ ) while  $\ell(0) = 1 \neq 0$ . The faithful statement adds the hypothesis  $x \neq 0$  (and the field hypotheses used by the Artin–Schreier telescoping). Likewise `eqn1_nonzero_root_unique` counts the *nonzero* solutions of eqn (1), because  $x = 0$  always solves the cleared equation; this requires  $c \neq 0$ . Both corrections are documented in the sources.

### 7.3 The root count (two cases)



## 8 What the whole development reduces to

Stripped to the non-trivial mathematical inputs, the entire library bottoms out on a small number of Mathlib results:

- `FiniteField.pow_card` (Fermat) and `iterateFrobenius_inj` — the Frobenius engine;
- `add_pow_char_pow` (freshman's dream) — all char-2 telescoping;

- `Nat.pow_sub_one_gcd_pow_sub_one` — the Mersenne-gcd fact;
- `Finite.injective_iff_surjective` — pigeonhole “injective  $\Rightarrow$  permutation”;
- `Set.ncard_eq_one` — “exactly one solution”.

The routine char-2 algebra on top of these is discharged by the general tactics (`ring`, `simp`, `grind`, `decide`, `omega`).

## 9 Simplifications made in the refactoring

- **Merged the trace theory.** The scattered trace lemmas (freshman’s dream, telescoping, Frobenius invariance, Artin–Schreier vanishing, idempotence/bit) now live in one `FiniteField/Trace` module, under one namespace, so a consumer imports a single coherent theory.
- **Split the prerequisites by concept.** Frobenius facts and trace facts, previously in one `FiniteFieldPrereqs` file, are now separate, independently reusable modules.
- **Isolated the number theory.** `mersenne_coprime/inv_mod_exists` are field-agnostic and now sit in their own `NumberTheory` module, ready to upstream.
- **Named declarations after their content.** Paper-numbered names (`theorem_5`, `theorem_1`, ...) became statement-describing names (`qeps_bijective_iff`, `qKasami_bijective_iff`, ...); the `gmap` definition now lives with the criterion that uses it.
- **Tidied proofs.** Removed a leftover `exact?`, dropped unused section variables and `simp` arguments. No proof was weakened and the axiom footprint is unchanged.